

Choosing Models and Solution Approaches

ML Model and Foundation Model Development

01

**THEORY-ONLY
LEARNING SERIES**

Model choices • Training • Evaluation



Every model choice starts with the problem

A useful model must fit the task, the data, and the operating limits.

Business task

Prediction, classification, generation, search, summarization, or extraction

Data conditions

Volume, labels, modality, language, quality, and freshness

Operating limits

Latency, cost, explainability, privacy, and maintenance



Traditional ML models solve different patterns

The learning setup narrows the possible algorithm families.

Supervised learning

Uses labeled examples for classification or regression.

Unsupervised learning

Finds clusters, structure, or unusual behavior without target labels.

Time series and recommendations

Use order, history, context, and repeated behavior.



Foundation models require a different selection process

Amazon Bedrock provides access to multiple foundation-model families.

Capability fit

Check supported tasks, languages, context length, and modalities.

Quality fit

Compare responses on examples that represent the real use case.

Operational fit

Review latency, throughput, customization options, and price.



Build, buy, or adapt

The best approach balances control with delivery time and operational effort.

More customization

- Custom algorithms or training
- Greater control over behavior
- More data and ML expertise
- Higher maintenance effort

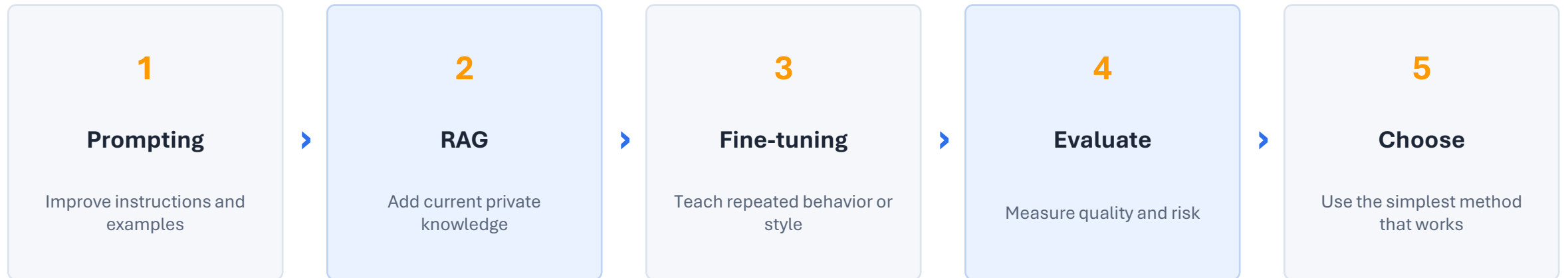
More management

- AWS AI services or pretrained FMs
- Faster implementation
- Less infrastructure work
- Behavior may be less flexible



Prompting, RAG, and fine-tuning solve different needs

Customization should match the type of gap in the base model.



AWS AI services can replace custom modeling

Managed services fit common business tasks with limited training effort.

Documents and text

Amazon Textract extracts document content. Amazon Comprehend analyzes language.

Images and video

Amazon Rekognition detects visual content and supports image analysis.

Speech

Amazon Transcribe converts speech to text for later analysis.



A repeatable decision checklist

Model selection depends on evidence, not popularity.

- 1 Define the output and acceptable errors.
- 2 Identify available data, labels, and modalities.
- 3 Compare model quality on representative examples.
- 4 Estimate training time, inference latency, and cost.
- 5 Check explainability, privacy, safety, and maintenance.
- 6 Select the simplest approach that meets the requirement.

